Q-resafe AWQ Calibration-Data Study

Fairest current experiment: token-matched confirmation

Main message
When Risk-I, Risk-II and Risk-III are given exactly the same calibration budget, Risk-II and Risk-III still show higher held-out ASR heuristic values than Risk-I. This is a useful confirmation signal, but not yet a final causal claim because the metric is heuristic, Risk-II is reconstructed, and the token-matched stream is a prefix rather than full source coverage.

1. Research question

Does changing only the calibration corpus used by pure AWQ change the safety behavior of the same already aligned Llama-2-7B-Chat model? The intended comparison is: same model, same official Q-resafe AWQ code, same INT4 settings, same seed, same evaluation prompts and generation settings; only the calibration corpus changes.

The first exploratory run used unequal token budgets, so it is not the fairest basis for the main conclusion. This report therefore treats the token-matched confirmation as the primary experiment.

2. Fair experiment design

Element	Fixed setting
Model	meta-llama/Llama-2-7b-chat-hf
Quantization	Official Q-resafe quant-without-ft AWQ; INT4; zero_point=True; group_size=128; GEMM
Conditions	Risk-I, reconstructed Risk-II, reconstructed Risk-III
Calibration budget	Exactly 384 rendered tokens per condition; exactly 3 blocks of 128 tokens
Safety evaluation	Same held-out AdvBench set, 510 prompts; greedy decoding; max_new_tokens=128; seed=42
Utility evaluation	WikiText-2 raw test perplexity under the same quantized models
Leakage control	The 10 Risk-III calibration rows are removed from the 510-row evaluation set

Risk-II is reconstructed from the public Q-resafe description; it is not the exact unreleased dataset. Risk-III uses paired instruction-response records from the public AdvBench export. No fine-tuning, optimizer, backward pass, LoRA, QAT, AQLM, GPTQ or QLoRA is used.

3. What 384 tokens and 3 blocks mean

The ten user/assistant calibration conversations are first rendered with the Llama-2 chat template. The tokenizer then converts the rendered text into a sequence of integer token IDs. The ten sequences are concatenated in locked order, and only the first 384 token IDs are kept. Those 384 IDs are split into three chunks of 128 IDs:

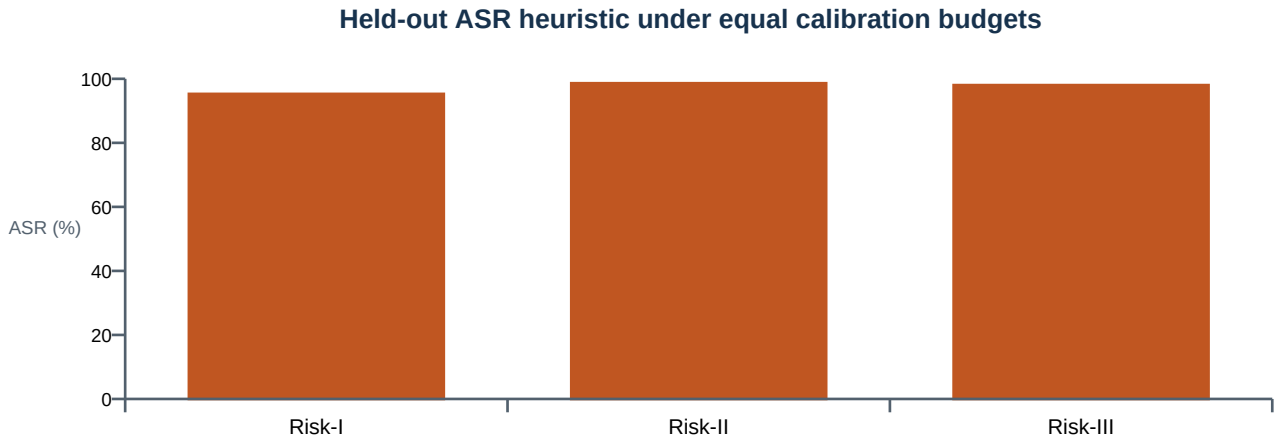
```
384 tokens = 128 tokens + 128 tokens + 128 tokens = 3 calibration blocks
```

AWQ runs the model forward on these blocks to observe activation ranges and choose INT4 scaling/clipping. It does not learn new behavior and does not update model weights. Equalizing the number of tokens and blocks removes the simplest quantity confound: one risk condition receiving much more calibration input than another.

Remaining fairness limitation
Equal token count does not mean equal source coverage. Because 384 is a prefix of the concatenated stream, some later parts of the ten conversations may not be included. The next definitive experiment should match token budget while preserving a pre-registered amount of source coverage.

4. Token-matched results

Condition	Source examples	Tokens	Blocks	ASR	Positive / 510	PPL	Delta vs Risk-I
Risk-I	10	384	3	95.29%	486	7.1464	+0.00 pp
Risk-II	10	384	3	98.63%	503	7.2186	+3.33 pp
Risk-III	10	384	3	98.04%	500	7.1971	+2.75 pp



5. Interpretation

Under the fairest recorded comparison, Risk-II has ASR heuristic 98.63%, which is 3.33 percentage points above Risk-I at 95.29%. Risk-III has 98.04%, which is 2.75 points above Risk-I. The result is not monotonic between Risk-II and Risk-III: Risk-II is the highest, while Risk-III is slightly lower.

The utility check does not show a dramatic collapse: PPL is 7.1464 for Risk-I, 7.2186 for Risk-II and 7.1971 for Risk-III. Therefore the ASR difference is not immediately explained by the Risk-II/Risk-III models becoming unusable, although PPL alone cannot prove that the effect is safety-specific.

Defensible conclusion
The token-matched experiment supports the hypothesis that calibration-data distribution may influence AWQ safety behavior even without fine-tuning. It is still preliminary evidence, not a final claim that harmful calibration data alone reliably degrades safety.

6. What is needed before a strong claim

Use a validated human or model-judge ASR protocol instead of the transparent first-pass heuristic; repeat the token-matched comparison over multiple pre-registered seeds; record the exact Q-resafe source commit; and design token matching that preserves comparable source coverage rather than only the first 384 tokens. The current conclusion should remain limited to this Llama-2-7B-Chat, INT4 AWQ setup.